ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Formation of new substance indicates chemical change.
2	**(d)** 44 g	Mass is conserved: 12 + 32 = 44 g.
3	**(b)** copper	CuO is reduced to Cu by hydrogen.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Energy is needed to break bonds.
5	**(a)** 16 g	40 - 24 = 16 g.
6	**(d)** zinc	Zn is more reactive; $ZnSO_4$ is colourless.
7	**(c)** A is true but R is false.	A is true; R is false because nitrogen is unreactive.
8	**(b)** 36 g	4 + 32 = 36 g.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$2\,Mg + O_2 \longrightarrow 2\,MgO$: two reactants give one product, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2\,NaCl$; both true, but the white precipitate is due to the insolubility of $BaSO_4$, not because the reaction is called double displacement.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass of CO_2 = 12 + 32 = 44 g (1); Law of conservation of mass (1).	1+1
12	Gases NO_2 and O_2 escape (1); Only solid PbO remains; the total mass is still conserved (1).	1+1
13	$ZnCO_3 \longrightarrow ZnO + CO_2$ (1); Carbon dioxide (turns lime water milky) (1).	1+1
14	Iron (56) dissolves and copper (63.5) is deposited (1); Copper atoms are heavier, so the net mass rises (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Equation already balanced: $PbO + C \longrightarrow Pb + CO$ (1); PbO is the oxidising agent (1); C is the reducing agent (1).	1+1+1
16	Basic copper carbonate (1); Copper reacts with moisture, CO_2 and oxygen (1); Prevent by lacquer/paint or keeping dry (1).	1+1+1
17	Hydrogen gas (1); Burns with a pop sound near a flame (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1).	1+1+1
	OR: $Fe_3O_4 + 4\,H_2 \longrightarrow 3\,Fe + 4\,H_2O$ (2); H_2 is oxidised (1).	2+1
18	X is magnesium, Y is magnesium oxide (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Y is basic as $Mg(OH)_2$ forms with water (1).	1+1+1
	OR: Brown copper turns black: $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1); Black CuO turns brown again (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Reddish brown (the solid is Fe_2O_3) (1). (ii) Thermal decomposition (1). (iii) $2\,FeSO_4 \xrightarrow{\text{heat}} Fe_2O_3 + SO_2 + SO_3$ (1). (iv) Sulphur dioxide and sulphur trioxide (1); endothermic, because heat has to be supplied continuously for the reaction to proceed (1).	1+1+1+2
20	Zinc: bubbles of hydrogen (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Copper: no reaction (1); Zinc is above hydrogen in the reactivity series (1); Copper is below hydrogen so cannot displace it (1).	1+1+1+1+1
	OR: (i) Zn is more reactive (1); (ii) Copper is deposited on the nail (1); (iii) Silver is formed by photolysis (1); (iv) Basic copper carbonate forms in moist air (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$; $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; H_2O must not be written as H_2O_2 to balance oxygen.
- Writing $Mg + O_2 \longrightarrow MgO$ without balancing, or MgO_2 for magnesium oxide.

- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $\mathrm{Cu} + \mathrm{FeSO_4}$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 40 mL	Ratio H2 : O2 = 2 : 1.
2	**(b)** increases slightly	Heavier Cu (63.5) replaces Fe (56) on the nail.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Reactivity decides displacement.
4	**(d)** 23	Al2(SO4)3: 2+3+12 = 17; 3H2: 6; total 23.
5	**(d)** hydrogen	Hydrogen explodes with a pop.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Photolysis gives grey silver.
7	**(d)** 12	$3\,Ca(OH)_2 + 2\,H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + 6\,H_2O$: 3+2+1+6=12.
8	**(c)** tube (iii) only	Needs both air and water.
9	**(c)** A is true but R is false.	Iron gains oxygen (oxidation), so A is true; iron is oxidised, not reduced, so R is false.
10	**(d)** A is false but R is true.	$CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$ absorbs heat (endothermic), so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	H_2S loses hydrogen: oxidised (1); Cl_2 gains hydrogen: reduced (1).	1+1
12	$CaO + H_2O \longrightarrow Ca(OH)_2$ (1); $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (1).	1+1
13	Correct formulae (1); $Na_2SO_4\,(aq) + BaCl_2\,(aq) \longrightarrow BaSO_4\,(s) + 2\,NaCl\,(aq)$ (1).	1+1
14	Silver reacts with hydrogen sulphide in air (1); Silver sulphide (Ag_2S) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2\,Mg + SO_2 \longrightarrow 2\,MgO + S$ (2); SO_2 is reduced (loses oxygen) (1).	2+1
16	X is quick lime, CaO; $CaCO_3 \longrightarrow CaO + CO_2$ (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Y is slaked lime, an exothermic product used for whitewash (1).	1+1+1
17	White precipitate of AgCl (1); $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$ (1); Double displacement (precipitation) (1).	1+1+1
	OR: $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$; 65 g Zn gives 2 g H2 (1); 6.5 g Zn = 0.1 mol (1); Mass of H2 = 0.2 g (1).	1+1+1
18	Low temperature slows oxidation (rancidity) (1); Antioxidants (1); Airtight packing / nitrogen flushing (1).	1+1+1
	OR: $2\,Mg + O_2 \longrightarrow 2\,MgO$: two elements give one compound (1); Mg gains oxygen: oxidised (1); O_2 is reduced (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Copper metal (1). (ii) Displacement reaction (1). (iii) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1). (iv) No reaction / no change (1), because copper is less reactive than iron and cannot displace iron from its salt solution (1).	1+1+1+2
20	$CaCO_3 \longrightarrow CaO + CO_2$ (1); CaO = 56 kg (1); CO2 = 44 kg (1); Thermal decomposition (1); Quick lime used in making cement/whitewash (1).	1+1+1+1+1
	OR: Flask weighed with both solutions kept apart (1); Mix them by tilting (1); Yellow precipitate of PbI_2 forms (1); Total mass is unchanged (1); Law of conservation of mass is verified (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; H_2O must not be written as H_2O_2 to balance oxygen.
- Writing $Mg + O_2 \longrightarrow MgO$ without balancing, or MgO_2 for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $Cu + FeSO_4$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.

- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Cu^{2+}	Cu2+ gains electrons/ is converted to Cu.
2	**(c)** A is true but R is false.	A true; R false because there is no oxidation or reduction.
3	**(d)** 3 g	Total mass 15 g; 15 - 12 = 3 g.
4	**(b)** magnesium or zinc	More reactive metals corrode in preference.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but the ratio is explained by formula H_2O, not merely being a compound.
6	**(d)** 1.1 g	50.0 - 48.9 = 1.1 g.
7	**(d)** changed the formula of the product, which is not allowed	Balancing is done by coefficients only, not by altering formulae.
8	**(d)** A is false but R is true.	Mg is oxidised, so A is false; R is true.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$; blue Cu^{2+} is replaced by colourless Zn^{2+}.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the 2 : 1 ratio comes from the formula H_2O ($2\,H_2O \longrightarrow 2\,H_2 + O_2$), which R does not state.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Exothermic reaction (heat is released) (1); e.g. $CaO + H_2O \longrightarrow Ca(OH)_2$ (1).	1+1
12	$2\,Mg + CO_2 \longrightarrow 2\,MgO + C$ (1); Mg is oxidised (CO_2 is reduced) (1).	1+1
13	Iron is more reactive and displaces copper (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1).	1+1
14	Volume of hydrogen = 20 mL (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ gives 2 : 1 by volume (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	100 g gives 56 g CaO and 44 g CO2 (1); CaO = 5.6 g (1); CO2 = 4.4 g (1).	1+1+1
16	Zinc is more reactive than copper (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); A less reactive metal cannot displace a more reactive one (1).	1+1+1
17	$CaO + H_2O \longrightarrow Ca(OH)_2$ releases heat: exothermic (1); $CaCO_3 \longrightarrow CaO + CO_2$ absorbs heat: endothermic (1); Two reactions are opposite in direction/energy (1).	1+1+1
	OR: Molar mass AgCl = 143.5 g; 2AgCl -> 2Ag + Cl2 (1); 14.35 g = 0.1 mol (1); Silver = 0.1 x 108 = 10.8 g (1).	1+1+1
18	Subscripts must not be changed in balancing (1); H_2O_2 is a different compound (hydrogen peroxide) (1); Correct: $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ (1).	1+1+1
	OR: Oxygen oxidises iron (1); Moisture speeds up rusting / provides electrolyte (1); Zinc forms a protective layer and corrodes first (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Rancidity (1). (ii) Nitrogen is unreactive and prevents oxidation of the fats and oils in chips (1). (iii) Oxygen (air) and moisture (water) (1). (iv) Salt in the moist sea air speeds up corrosion (1); painting, galvanising (zinc coating) or greasing prevents rusting (1).	1+1+1+2
20	(a) $4\,Fe + 3\,O_2 \longrightarrow 2\,Fe_2O_3$, combination (2); (b) $2\,NH_3 + 3\,CuO \longrightarrow 3\,Cu + N_2 + 3\,H_2O$, redox (1); (c) $8\,Al + 3\,Fe_3O_4 \longrightarrow 4\,Al_2O_3 + 9\,Fe$, displacement/redox (2).	2+1+2
	OR: Thermal: $CaCO_3 \longrightarrow CaO + CO_2$ (heat) (1); Electrolytic: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (electricity) (1); Photolytic: $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (light) (1); Application of photolytic: photography (1); Application of electrolytic: extraction of hydrogen/oxygen (1).	1+1+1+1+1

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